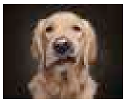
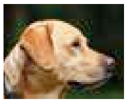


Social cognition in dogs
During the process of domestication, dogs develop social cognition to communicate with humans. **http://dgit.in/mar21-07**



Decoding dog domestication
Advances in the environmental adaptation of dogs provide new perspectives on human & dog communication. **http://dgit.in/mar21-08**

levels of adrenaline in the domesticated foxes. The coloration of the foxes also began to vary a lot, which led Belyayev to suspect that adrenaline shares a biochemical pathway with melanin, a theory that has since then been confirmed. Researchers have studied the brains of the domesticated silver fox, and found changes in the regions of the brain related to emotions and social behavior, exactly the places where the researchers expected to see them. The current research is further investigating the neurochemistry of these domesticated foxes, investigating the roles of vasopressin and serotonin, both of which are linked to aggression, and oxytocin which is related to social bonding.

While the selection for tameness remains a popular theory, the findings have been disputed. Some researchers believe that it was the dogs that made the choice of domesticating themselves, and not humans. Opportunistic wolves might have followed bands of humans as they helped them survive. Other researchers believe that less timid wolves might have been the guiding principle for early domestication. The more curious and friendly wolves developed the social skills for communicating with humans, leading to the early proto dogs.

THE TALE IN THE BONES

Paleoecologists and bioarchelogists studying ancient dog bones found in the sites of ancient human settlements, have traced the domestication of dogs to the surplus of meat among humans bands in the last ice age. The research was conducted by scientists from Finnish Food Authority, and published in Nature in January this year. According to this study, the main factor in the domestication of dogs was that humans were not entirely carnivorous species. The large animals in the harsh environment of the ice age had little to eat, and their bodies had little fat. Most of their bodies were made of lean meat, which humans cannot eat very much of without getting ill. Humans could derive nourishment from the bone marrow and the fat and grease around it, but the excess lean meat from the hunts would have been discarded away. Wolves can stomach more lean meat, and thus could have easily gobbled up the surplus among the human bands. It was these early wolves that did not need to compete with humans, that became friendly and domesticated.

THE STORY IN THE GENES

Evolutionary geneticists and archaeogeneticists have been studying the genomes of various breeds of existing dogs, DNA retrieved from remains of ancient dogs from burial sites, and various species of wolves, in the hopes that comparing the DNA will provide insights into when and where dogs diverged from wolves. This is an area of active research, and there is considerable disagreement on what the DNA tells us.

A domestic silver fox

For example, research that compared the genomes of chinese dog breeds, European dog breeds and grey wolves, came to the conclusion that the dogs were domesticated in China around 32,000 years ago. The problem with the finding is that there were no wolves in China that could possibly get domesticated at that time. The findings of other studies paint a far more complex picture, with domestication occurring between 16,000 and 11,000 years ago, and that both populations of dogs continued to interbreed with the wolves long after domestication. Another study painted a similarly murky picture, that dogs were domesticated twice, and that an older population of European dogs had been wiped out by an immigrant population of dogs from Asia during the neolithic era, which started around 12,000 years ago. Another study conducted on some of the oldest dog remains, from 15,000 years ago, found in Germany suggests that there was a single domestication event at some point between 40,000 and 20,000 years ago. The most recent study sequenced the genomes of 27 ancient dogs ranging from 100 to 11,000 years old, ranging as far wide as Siberia, the Middle East and Europe. The study found that ancient dog populations moved with ancient human populations. Farmers from the Middle East moving into Europe took their dogs with them. The Middle Eastern dogs from around 7,000 years ago are genetically linked to modern dog populations in Africa, potentially signs of dog populations moving with humans in a back migration to Africa that took place around that time. However, there is no evidence of dog populations changing because of a human migration from the Russian steppes into Europe that took place around 5,000 years ago. The study also noted the times when the populations of dogs appeared to change, but not the people. The findings pose more questions than answers, with the mystery of dog domestication only deepening the more researchers study it. The researchers refrained from pinpointing a date for the domestication in the study, but showed that there were already at least 5 varieties of dogs as early as 11,000 years ago. To paint a more complete picture it is necessary to sequence and study the genomes of more ancient dogs and wolves.

The brains and the DNA of dogs have changed and evolved over time and all the research into the ancient and modern dogs hints at the similar changes that humans have gone through, when early humans domesticated themselves. d